Think

When a fixed amount is deposited for a fixed rate of interest, the simple interest changes propotionaly with the number of years it is being deposited. Can you find any other examples of such kind.

Example 4.1

If 6 children shared 24 pencils equally, then how many pencils are required for 18 children?



Solution

Let x be the number of pencils required for 18 children. As the number of children increases, number of pencils also increases.

In the case of direct proportion we take $\frac{x_1}{y_1} = \frac{x_2}{y_2}$

Number of children	6	18
Number of pencils	24	x

$$\frac{6}{24} = \frac{18}{x}$$

$$\frac{6 \times x}{24} = 18$$

$$x = \frac{18 \times 24}{6} = 72$$

Hence, 72 pencils are required for 18 children.

Example 4.2

If 15 chart papers together weigh 50 grams, how many of the same type will be there in a pack of $2\frac{1}{2}$ kilogram?

Solution

Let x be the required number of charts.

Number of chart papers	15	x
Weight in grams	50	2500

As weight increases, the number of charts also increases. So the quantities are in direct proportion.

Hence $\frac{x_1}{y_1} = \frac{x_2}{y_2}$



$$\frac{15}{50} = \frac{x}{2500}$$

$$15 \times 2500 = x \times 50$$

$$x \times 50 = 15 \times 2500$$

$$x = \frac{15 \times 2500}{50} = 750$$

Therefore, 750 charts will weigh $2\frac{1}{2}$ kilogram.

Unitary Method

We know the concept of unitary method studied in the previous class.

First, the value of one unit will be found. It will be useful to find the value of the required number of units. We will try to solve more problems using unitary method.

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